

Dominic Nguyen

Vancouver, BC

(604)-716-3930 | DominicNguyen817@gmail.com | www.linkedin.com/in/dominicn17

TECHNICAL SKILLS

Computer: SOLIDWORKS, C, C#, MATLAB, Microsoft Excel, KiCAD, Python, Windows Forms

Laboratory: Soldering, Oscilloscope, Digital Multimeter, Circuit Analysis, 3D Printing

Machine Tools: Bandsaw, Milling Machine, Lathe, Drill Press, Press Brake

TECHNICAL PROJECTS: [PROJECT PORTFOLIO](#)

Litter Collecting Drone Capstone, UBC

October 2025 – April 2026

- Rapidly prototyped various claw designs for collecting lightweight litter commonly found on hiking trails, 3D printing and assembling to test claw design effectiveness.
- Designed a claw mechanism using herringbone gears and tight tolerances to eliminate backlash, maximizing range of claw usability.
- Set up flight tests to measure minimum hovering height for pickup task

Biped Wheeled Robot Pair Project, UBC

December 2025 – February 2026

- Designed circuit schematic diagram, choosing appropriately rated components to ensure project success.
- Coded firmware to estimate tilt angle using a Madgwick filter with an MPU-6500 accelerometer and gyroscope.
- Deployed a Madgwick filter to minimize noise from external acceleration while maintaining a bandwidth of 10Hz.
- Applied multiprocessing on an ESP32 using both CPUs, improving processing speed and ensuring precise timing of tilt estimation at 1 kHz and PID calculation loop at 500 Hz.
- Decorruped IMU data by switching communication protocol from I2C to SPI, resisting motor power and signal noise and allowing for accurate usable sensor data.

25:1 Gear Ratio Cycloidal Reducer, Personal Project

July 2025 – August 2025

- Maximized gearbox efficiency by choosing low friction materials and lubricating contact points, yielding an efficiency of 89%.
- Applied parametric equation in SOLIDWORKS to efficiently design cycloidal rotor, saving hours of time.
- Eliminated backlash by designing to 3D printer tolerances, producing tolerances of less than 0.02mm.
- Experimentally identified a skew error of 0.24% on the 3D printer axis and generated a python script to shift “gcode” instructions, eliminating mechanical skew error.

Quadruped Spider Robot, Personal Project

April 2025 – May 2025

- Designed and dimensioned components using SOLIDWORKS, aligning screw holes and minimizing material use.
 - Deployed inverse kinematics on spider legs with 2 degrees of freedom, optimizing smoothness of walking.
 - Optimized manufacturability by designing flat components for 3D printing, resulting in high quality components.
 - Prototyped circuit by designing a circuit schematic using KiCAD and testing on breadboard, minimizing risk.
 - Designed component symbols on KiCAD, allowing more legible circuit documentation.
-

EDUCATION

University of British Columbia

Bachelor of Applied Science - Mechanical Engineering, Mechatronics Option

PROFESSIONAL AFFILIATIONS

EGBC - Engineers and Geoscientists British Columbia

2023 – Present